Utku Kalkanli

MLOps Engineer

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Available immediately · Full-time

MLOps engineer with three years at Linde taking ML models from data to production: Airflow pipelines, GPU serving with TensorRT and Triton, Azure ML, and monitoring.

Experience

Linde GmbH, Digitalization Oct 2023 – Sep 2026
Machine Learning Engineer Munich, Germany

- Maintained and optimized a production Azure ML pipeline that forecasts CPAP therapy adherence for ∼1M patients daily. Cut runtime from 37 to 8 hours (78%) with modular training/inference and incremental ETL.
- Built monitoring for data quality, drift, and forecast performance, with alerts raised as Azure DevOps work items and secrets in Azure Key Vault.
- Supported the pipeline's EU MDR certification as medical device software: technical documentation, IEC 62304 lifecycle documents, risk analysis, validation and test evidence, traceability, and Azure cybersecurity testing.
- Built end-to-end Airflow DAGs for YOLO and RF-DETR (data preparation, training, ONNX/TensorRT export, Triton benchmarking) on a Docker Compose stack (Airflow, MLflow, PostgreSQL, Redis, GPU workers).
- Delivered an RF-DETR small-object detector for satellite imagery (mAP@50 0.66 → 0.83) that fully automated an industrial asset-monitoring process; deployed for the sales team.
- Contributed to a rotor-orbit fault classifier deployed at a Linde air separation plant, where it predicted a real equipment failure. Named inventor on the invention; co-author of an internal predictive-maintenance whitepaper.

Fraunhofer IKS Oct 2025 – Oct 2026
Research Assistant, Machine Learning (Master's Thesis) Munich, Germany

- Ran distributed GPU experiment pipelines with Ray on a Slurm cluster, versioned in GitLab, for multi-view 2D/3D anomaly detection (diffusion-based synthetic data, feature banks, point-cloud registration).
- Building a LangGraph and Qwen tool-calling agent that orchestrates multi-stage PyTorch experiments and evaluation.

Novit.AI Mar 2022 – Jun 2022
Machine Learning Engineer, Internship Ankara, Türkiye

- Built and deployed a containerized edge AI system (FastAPI, Docker, XGBoost, GitHub Actions, NGINX) with real-time inference and automated rollout to 50+ NVIDIA Jetson devices.

Open Business Software Solutions; Cybersoft 2020 – 2021
Software Engineer, Internships Türkiye

- Built microservices and web apps with Python, Django, PostgreSQL, Elasticsearch, React and TypeScript on GCP.

Education

Technical University of Munich (TUM) Apr 2023 – Oct 2026
M.Sc. in Informatics Munich, Germany
Focus: machine learning, databases, high-performance computing. Thesis: *Reference-Guided Multi-view Anomaly Detection for Industrial Objects* (Prof. Stephan Günnemann, at Fraunhofer IKS).

Bilkent University Sep 2017 – Jun 2022
B.Sc. in Computer Science Ankara, Türkiye

Projects

Radar-Based Ship Identification (TUM industry project with Covadonga Maritime, Nov 2024 – Jun 2025)
Built an automated labeling pipeline matching radar sightings to AIS ship tracks (240 s temporal clustering; one year, 935k sightings, 388 ships), radar-signal features (peak detection, FFT, autocorrelation), and an XGBoost classifier: 91% accuracy on a class-balanced test set. Delivered to Covadonga.

Skills

MLOps: MLOps, Apache Airflow, MLflow, Model Registry, Model Monitoring, Drift Detection, CI/CD, Ray, Slurm, Celery, EU MDR, IEC 62304
Cloud: Microsoft Azure, Azure Machine Learning (Azure ML), Azure DevOps, Azure Key Vault, Azure Blob Storage, AWS (EC2, S3), GCP, GitLab CI, GitHub Actions
Containers and Data: Docker, Docker Compose, PostgreSQL, Redis, ETL, FastAPI, Microservices, NGINX, Linux
Model Serving: ONNX, TensorRT, NVIDIA Triton Inference Server, CUDA, NVIDIA Jetson, Edge AI
Machine Learning: Python, SQL, PyTorch, scikit-learn, Computer Vision, YOLO, RF-DETR, Hugging Face, XGBoost, Time-Series Forecasting
AI-Assisted Development: Claude Code, Codex, MCP
Languages: Turkish (native), English (C1, fluent), German (A2)